



BroadPOS Terminal Simulator

User Manual

[Version V2.00.02_20220309]

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Preface

BroadPOS Terminal Simulator_V2.00.02_20220309 User Manual

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Revision History

Date	Version	By	Description of Change
2011/09/21	V1.00.00_20110921	Jacob.W	Initial Version
2011/11/08	V1.00.01_20111108	Alfred.Z	Updated Version
2011/11/11	V1.00.01_20111111	Alfred.Z	Updated Version
2011/11/18	V1.00.01_20111118	Alfred.Z	Updated Version
2018/07/25	V1.00.12_20180725	Jerry.Z	Updated Version
2020/03/18	V2.00.00_20200318	Lenard.Z	Updated Version
2021/01/07	V2.00.01_20210107	Lenard.Z	Updated Version
2021/01/19	V2.00.01_20210119	Lenard.Z	Updated Version
2022/03/08	V2.00.02_20220309	Nathan.P	Updated Version

1 Introduction

1.1 Purpose

This user manual provides instructions and guidance on how to use the BroadPOS Terminal Simulator.

1.2 Acronyms

- **ECR:** Electronic Cash Register
- **POS:** Point of Sale

1.3 Audience

This document is intended for the following:

- System Developers of POS/ECR
- Users of Simulator: Sales team, authorized re-sellers, Technical Support, etc.

2 Background

2.1 BroadPOS Terminal Simulator

- BroadPOS Terminal Simulator is a tool to help developers develop a POSLink application.
- The Sales team, authorized re-sellers, and Technical Support can also use the BroadPOS Terminal Simulator to test and demonstrate SP30S, S300, and MT30S' features and functions.
- Use the BroadPOS Terminal Simulator to become more familiar with PAX Technology products.
- The tool simulates most POS terminal functions and acts as an actual POS terminal.

2.2 POS Apps

- BroadPOS Terminal Simulator software installation package includes three POS apps: **SP30S, S300, Px7, S920_Integration, S920_Standalone, D210S_Integration, D210S_Standalone** and **MT30S**.
- Tool users can also import other software packages, if necessary.

3 Apps Introduction

3.1 Start Apps

3.1.1 SP30S

To access the SP30S Simulator:

- Double click **SP30.exe** to run the BroadPOS Terminal Simulator.
- SP30S has serial port: **COM1**. To use this serial port, [configure UART ports](#) according to your PC's environment.
- If you want to change the **ECR communication type**, the operation steps are as follows: click the MENU button on the keyboard -> enter default password -> **6.Communication** -> enter default password -> **7.ECR Comm. Type**.

Figure: SP30S Simulator



3.1.2 MT30S

To access the MT30S Simulator:

- Double click **MT30.exe** to run the BroadPOS Terminal Simulator.
- MT30S has serial port: **COM2**. To use this serial port, [configure UART ports](#) according to your PC's environment.
- If you want to change the **ECR communication type**, the operation steps are as follows: click the ENTER button on the keyboard -> enter default password -> **6.Communication** -> enter default password -> the next page -> **2. ECR Comm. Type**.

Figure: MT30S Simulator

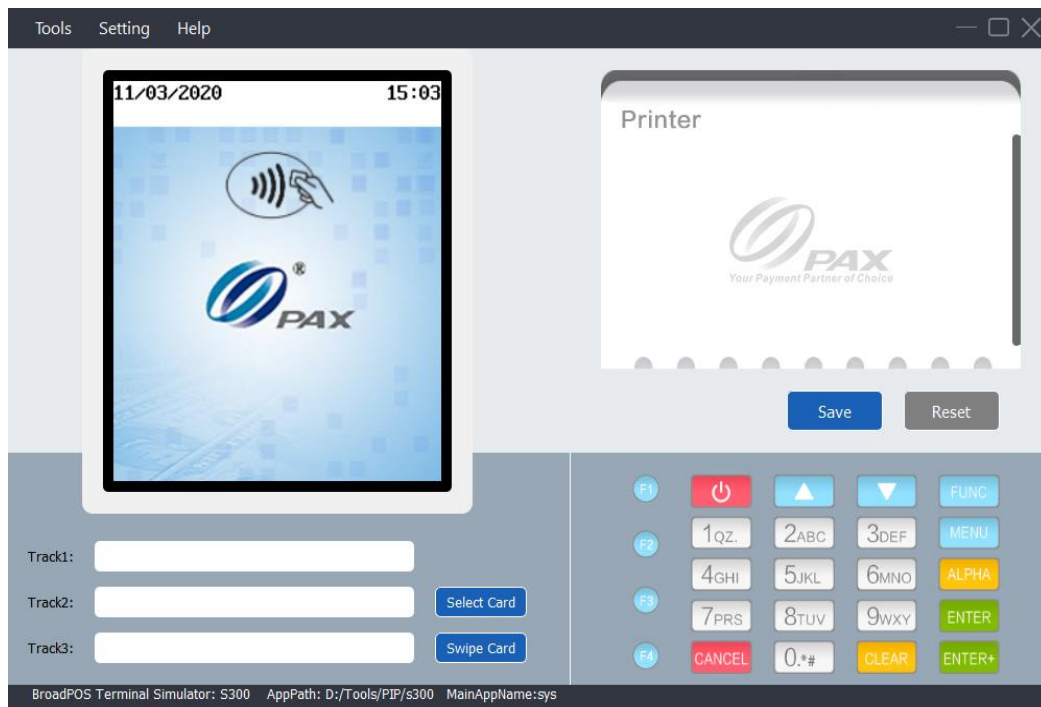


3.1.3 S300

To access the S300 Simulator:

- Double click **S300.exe** to run the BroadPOS Terminal Simulator.
- S300 has serial ports: **COM1**, **COM2**. To use these serial ports, [configure UART ports](#) according to your PC's environment.
- If you want to change the **ECR communication type**, the operation steps are as follows: click the F1 button on the keyboard -> enter default password -> click the button on the keyboard to indicate the next page -> **2.Communication** -> Enter default password -> click the button on the keyboard to indicate the next page -> **4.ECR Comm. Type**.

Figure: S300 Simulator



3.1.4 D210s_Integration

To access the D210s_Integration Simulator:

- Double click **D210s_Integration.exe** to run the BroadPOS Terminal Simulator.
- D210s has serial ports: **COM1**, **USBDEV**. To use these serial ports, [configure UART ports](#) according to your PC's environment.
- If you want to change the **ECR communication type**, the operation steps are as follows: click the MENU button in the keyboard -> enter default password -> click the button on the keyboard to indicate the next page -> **2.Communication** -> enter default password -> click the button on the keyboard to indicate the next page -> **3.ECR Comm. Type**.

Figure: D210s_Integration Simulator



3.1.5 D210s_Standalone

To access the D210s_Standalone Simulator:

- Double click **D210s_Standalone.exe** to run the BroadPOS Terminal Simulator.
- D210s has serial ports: **COM1**, **USBDEV**. To use these serial ports, [configure UART ports](#) according to your PC's environment.
- The model has no **ECR Communication Type**.

Figure: D210s_Standalone Simulator



3.1.6 PX7

To access the PX7 Simulator:

- Double click **PX7.exe** to run the BroadPOS Terminal Simulator.
- PX7 has serial ports: **COM2**, **USBDEV**. To use these serial ports, [configure UART ports](#) according to your PC's environment.
- If you want to change the **ECR communication type**, the operation steps are as follows: click F1 button on the keyboard -> enter default password -> **6.Communication** -> enter default password -> the next page -> **3. ECR Comm. Type** .

Figure: PX7 Simulator



3.1.7 S920_Integration

To access the S920_Integration Simulator:

- Double click **S920_Integration.exe** to run the BroadPOS Terminal Simulator.
- S920 has serial ports: **COM1**, **USBDEV**. To use these serial ports, [configure UART ports](#) according to your PC's environment.
- If you want to change the **ECR communication type**, the operation steps are as follows: click the MENU button in the keyboard -> enter default password -> click the button on the keyboard to indicate the next page -> **2.Communication** -> enter default password -> click the button on the keyboard to indicate the next page -> **4.ECR Comm. Type**.

Figure: S920_Integration Simulator

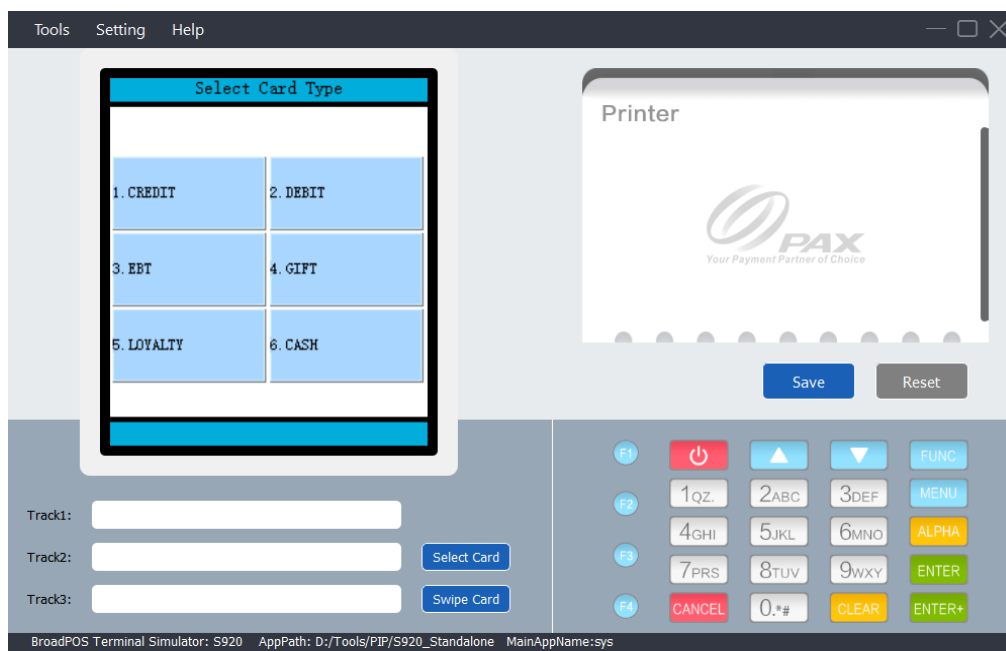


3.1.8 S920_Standalone

To access the S920_Standalone Simulator:

- Double click **S920_Standalone.exe** to run the BroadPOS Terminal Simulator.
- S920 has serial ports: **COM1**, **USBDEV**. To use these serial ports, [configure UART ports](#) according to your PC's environment.
- The model has no **ECR Communication Type**.

Figure: S920_Standalone Simulator



3.1.9 Other Apps

If you have other apps, you can import those by following steps:

- Access the **Resource** folder: double click **Simulator.exe** to run the BroadPOS Terminal Simulator.
- Click **Setting** menu > click **Import App**
- In the **Config App** pop up box > click **Browser** button and select applicable **App Directory**
- Click **Ok**
- **Warning** dialog box pops up: *The config will affect restarting! Do you want to restart now?*
- Click **OK**. Simulator automatically restarts.

4 Simulator Introduction

4.1 Simulator Main/Initial Screen



4.2 Menu Bar

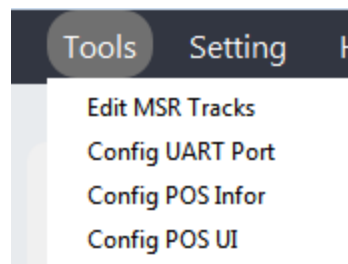
Menu toolbar includes:

- **Tools**
- **Setting**
- **Help**

4.2.1 Tools menu

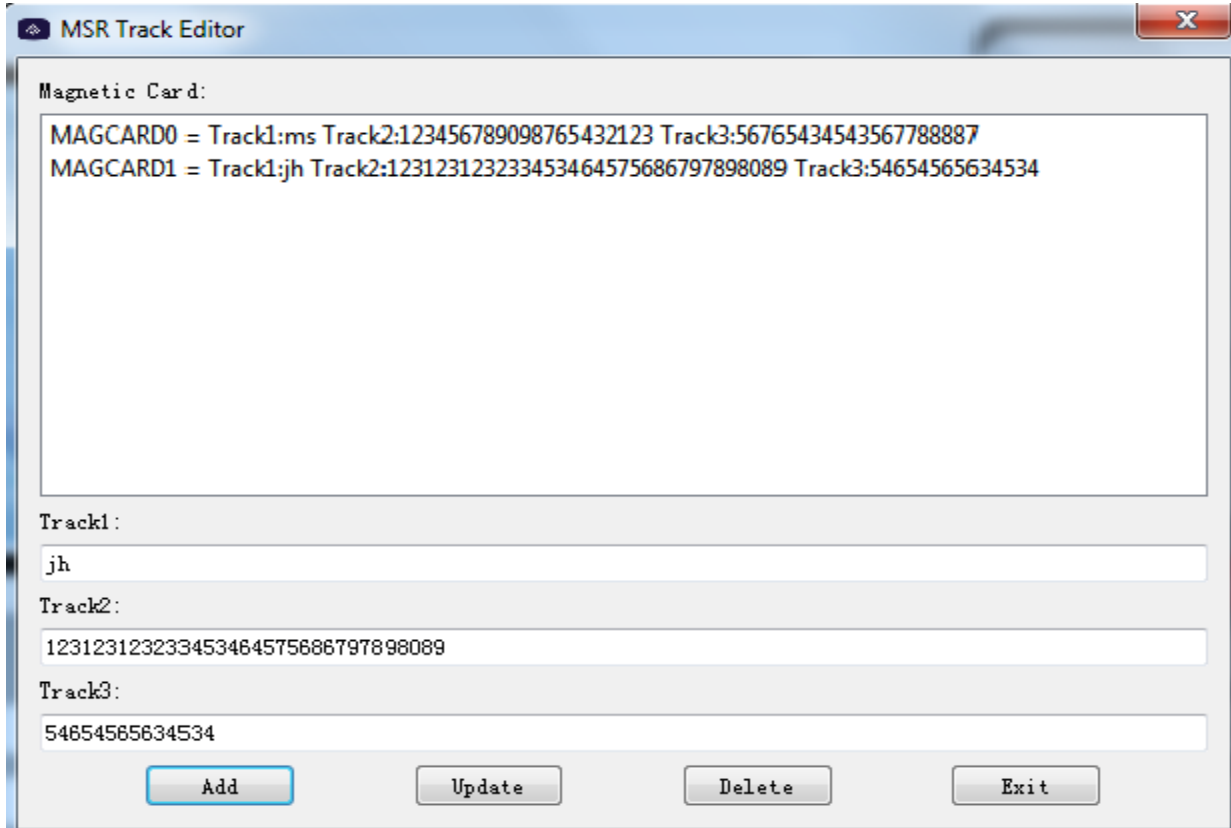
Tools menu includes four submenus:

- **Edit MSR Tracks**
- **Config UART Port**
- **Config POS Info**
- **Config POS UI**



4.2.1.1 Edit MSR Tracks

Figure: Edit MSR Tracks



The image shows a software window titled "MSR Track Editor". Inside, there is a section labeled "Magnetic Card:" containing two lines of text: "MAGCARD0 = Track1:ms Track2:123456789098765432123 Track3:56765434543567788887" and "MAGCARD1 = Track1:jh Track2:123123123233453464575686797898089 Track3:54654565634534". Below this, there are three input fields labeled "Track1:", "Track2:", and "Track3:". The "Track1:" field contains "jh", "Track2:" contains "123123123233453464575686797898089", and "Track3:" contains "54654565634534". At the bottom of the window, there are four buttons: "Add", "Update", "Delete", and "Exit".

Use the **Msr Track Editor** dialog box for managing and storing MagCard information. Use the stored card information instead of re-entering the information each time.

1. Utilizing **Menu** bar > Click **Tools**
2. Click **Edit Msr Tracks** to open **Msr Track Editor** dialog box

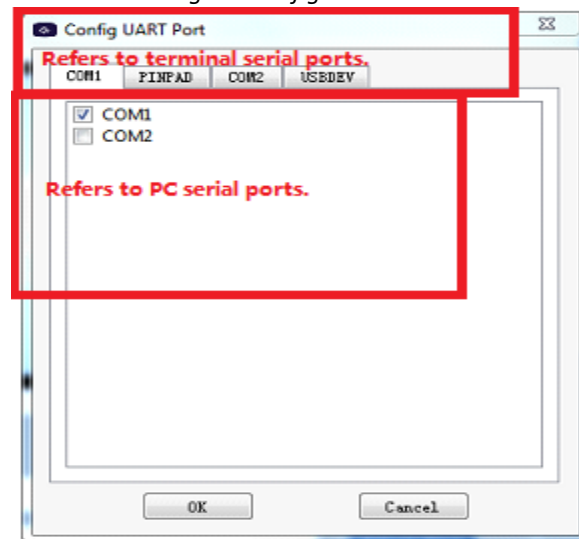
Add, Update, Delete, and Exit button functionality:

- A. **Add** button: use to add MagCard information. User cannot add duplicate MagCard information or empty value.
- B. **Update** button: use to modify MagCard information.
 - a. To update the MagCard information, select the applicable MagCard information in the **MagCard** field.
 - b. Click **Update** button to edit. Note: New MagCard values should not be empty or repeated.
- C. **Delete** button: use to delete MagCard information.

- a. To delete the MagCard information, select a MagCard information from the list above first.
 - b. Click **Delete** button
- D. **Exit** button: use to close **Msr Track Editor** dialog box.

4.2.1.2 Configure UART Port

Figure: Config UART Port



Use the **Config Uart Port** dialog box to configure serial ports in POS terminal. The configuration takes effect after restarting simulator.

1. Utilizing **Menu** bar > Click **Tools**
2. Click **Config Uart Port** to open **Config Uart Port** dialog box
3. Example: If POS terminal set to **COM1**: utilizing **COM1** tab > select applicable PC serial port
4. Click **OK** and simulator restarts automatically.

Config Uart Port Functionality:

- A. **COM1**, **PINPAD**, **COM2**, and **USBDEV** tabs are POS terminal serial ports.
- B. **COM1**, **COM2** are PC serial ports

Note:

- Different POS terminals have different serial ports. Configure according to the actual situation.
- In figure above, POS terminal **COM1** tab > select **COM1** to map the POS terminal serial port.

For example:

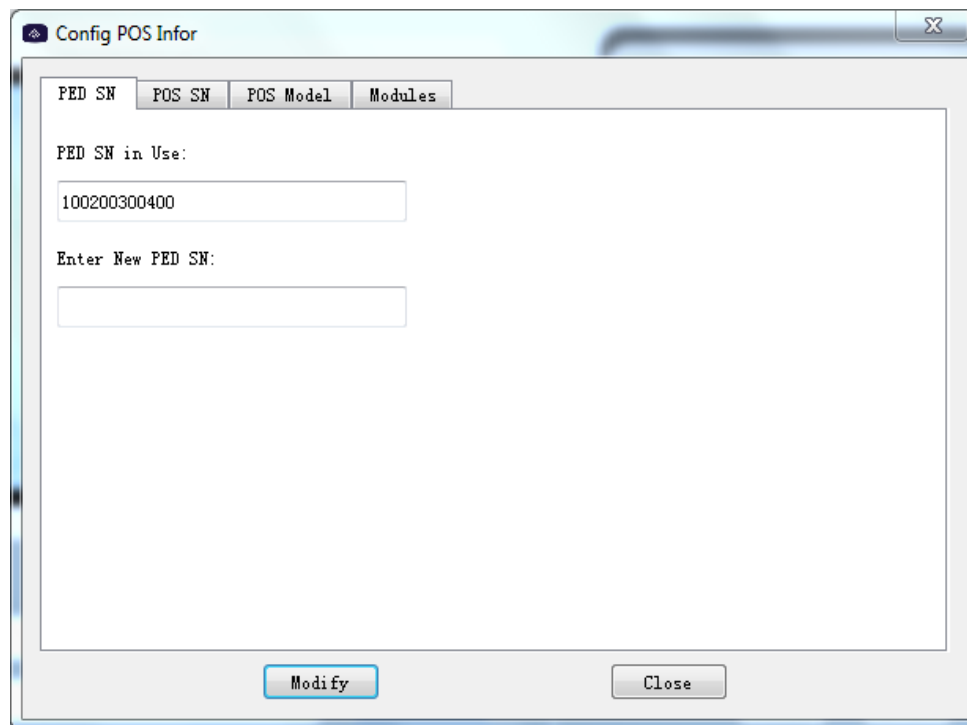
POSLink and Simulator are on a same PC: POS terminal has **COM1**, **COM1** and **COM2** are a pair of two virtual serial ports. Use the serial ports to make connection. Select **COM1** and restart Simulator. Select COM2 in POSLink.

4.2.1.3 Configure POS Info

1. Utilizing **Menu** bar > Click **Tools**
2. Click **Config POS Infor** to open **Config POS Infor** dialog box

4.2.1.3.1 PED SN tab

Figure: Configure PED SN

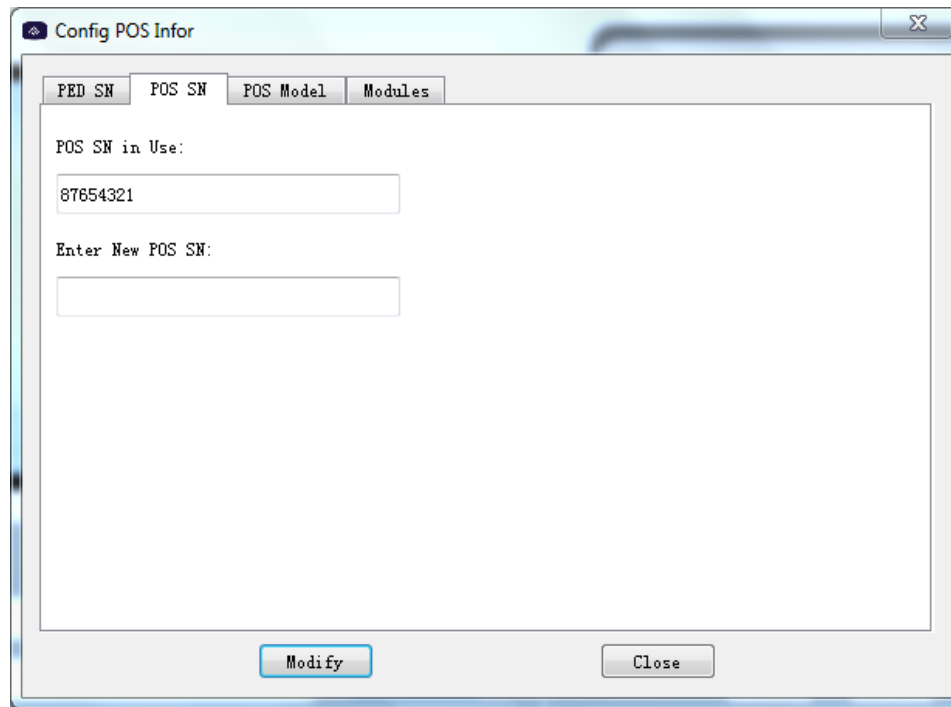


Use the **Config POS Infor** dialog box/**PED SN** tab to configure or modify a terminal's serial number. For simulation purposes, user may enter a "dummy" serial number. The configuration takes effect after restarting simulator.

1. **PED SN in Use** field: enter new PED SN
2. Click **Modify** .
3. After modifying all configurations, restart simulator.

4.2.1.3.2 POS SN tab

Figure: Configure POS SN



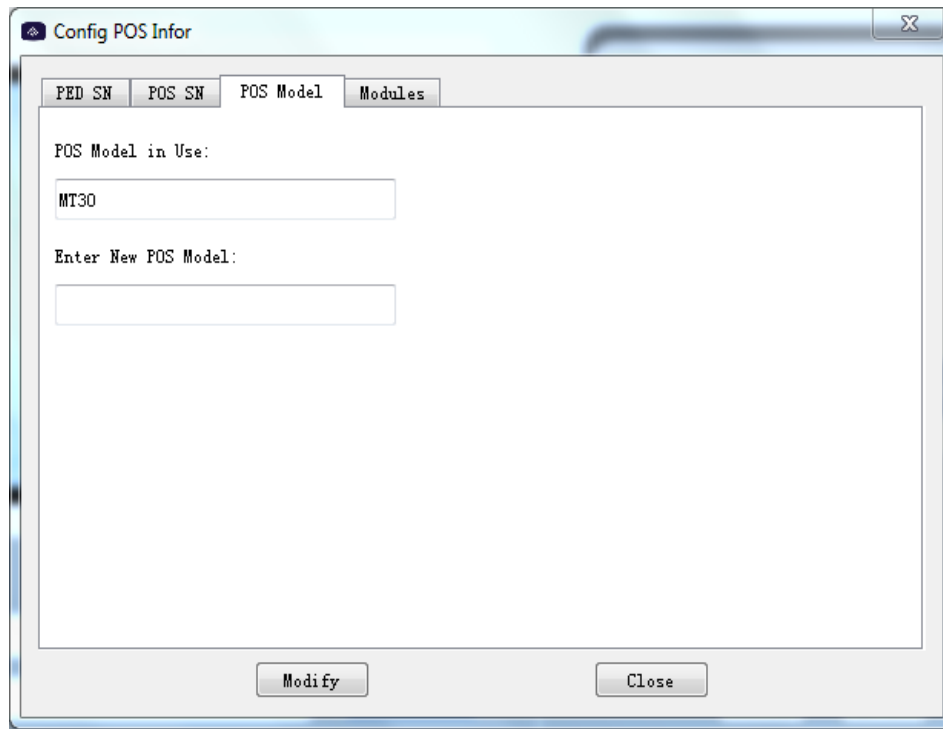
The image shows a software dialog box titled "Config POS Infor". It has four tabs: "PED SN", "POS SN", "POS Model", and "Modules". The "POS SN" tab is currently selected. Inside the dialog, there are two text input fields. The first field is labeled "POS SN in Use:" and contains the value "87654321". The second field is labeled "Enter New POS SN:" and is currently empty. At the bottom of the dialog, there are two buttons: "Modify" and "Close".

Use the **Config POS Infor** dialog box/**POS SN** tab to configure or modify a POS serial number. For simulation purposes, user may enter a “dummy” serial number. The configuration takes effect after restarting simulator.

1. **POS SN** field: enter new POS SN
2. Click **Modify**
3. After modifying all configurations, restart simulator.

4.2.1.3.3 POS Model tab

Figure: Configure POS Model



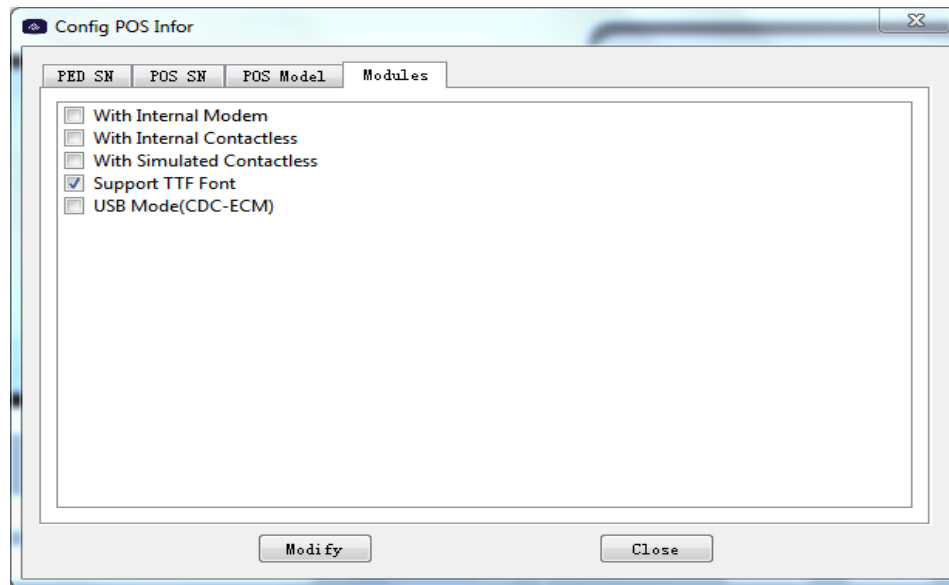
The screenshot shows a software window titled "Config POS Infor". It has four tabs: "PED SN", "POS SN", "POS Model" (which is selected), and "Modules". Inside the "POS Model" tab, there are two text input fields. The first field is labeled "POS Model in Use:" and contains the text "MT30". The second field is labeled "Enter New POS Model:" and is currently empty. At the bottom of the window, there are two buttons: "Modify" and "Close".

Use the **Config POS Infor** dialog box/**POS Model** tab to configure or modify a POS Model. The configuration takes effect after restarting simulator.

1. **POS Model in Use** field: enter new POS Model
2. Click **Modify**
3. After modifying all configurations, restart Simulator.

4.2.1.3.4 Modules tab

Figure: Configure Modules



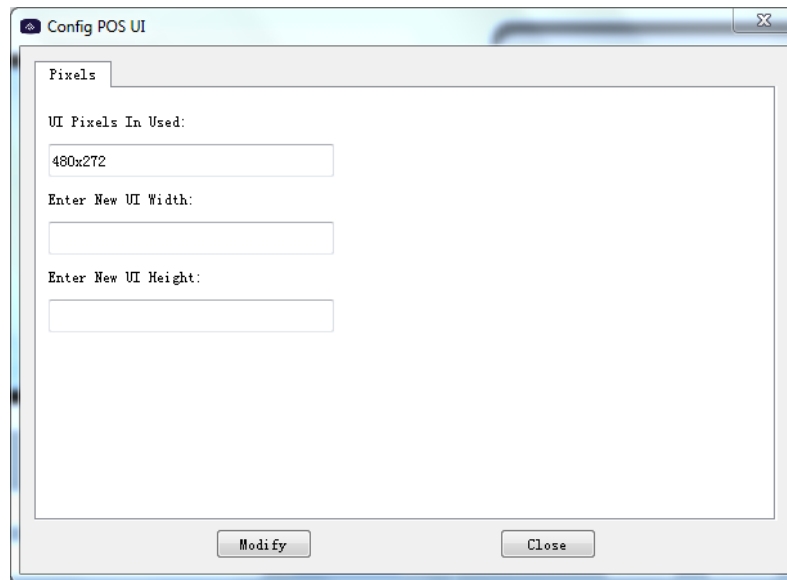
Use the Models tab to configure Modules. The configuration takes effect after restarting Simulator.

Modules Functionality:

- A. **With Internal Modem:** use for EM100.
 - a) Select this option and the Simulator simulates communication with the modem.
 - b) Connect EM100 to PC.
- B. **With Internal Contactless:** use for R50.
 - a) Select this option and the Simulator simulates communication with contactless module.
 - b) Connect R50 to PC.
- C. **With Simulated Contactless:**
 - a) Select this option and the Simulator simulates a contactless module.
 - When user needs to swipe contactless card, and app does not need to communicate with the contactless, use this option.
 - Some apps receive an error if they try to communicate with the contactless module; Simulator does not implement the communication protocol.
- D. **Support TTF Font:**
 - a) Select this option and Simulator uses TTF font.
 - The latest apps use TTF font.
 - Dot matrix font no longer meet the demand.
 - App packages do not contain font information, user needs to manually configure.
- E. **USB Mode(CDC-ECM)**
 - a) Whether to try this option on PC.

4.2.1.4 Configure POS UI

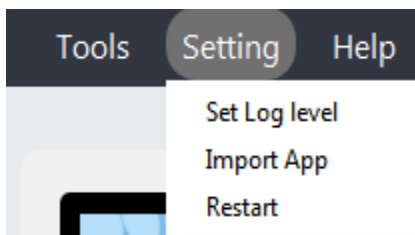
Figure: Configure POS UI



Use the **Config POS UI** dialog box to configure screen size. Enter POS terminal screen size. If the screen size configuration is incorrect, the screen area displays improperly.

1. **UI Pixels In Used** field: enter new screen size
2. **Enter New UI Width** field:
3. **Enter New UI Height** field:
4. Click **Modify**
5. After modifying all configurations, restart simulator.

4.2.2 Setting menu

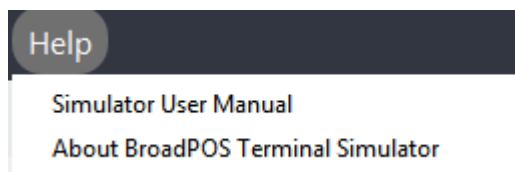


Setting Functionality:

- A. **Set Log level**: use to configure which level log to save.
 1. **Log Level** popup box > select applicable log level: **None**, **Critical**, **Warning**, or **All**
 2. Click **OK**
 3. **Simulator** popup box > Click **OK**
- B. **Import App**: select this function to specify which app to use.

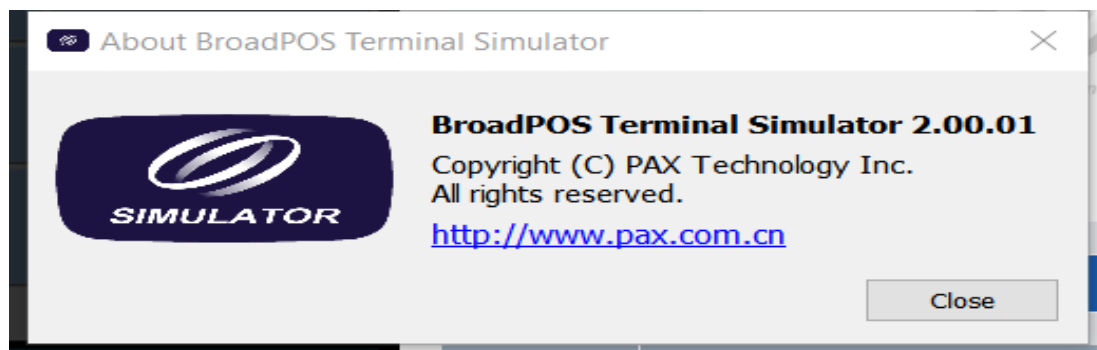
1. **Config App** popup box > click **Browser** > select app path
 2. **Sys Names** field: use drop down choose applicable name
 3. Click **OK**
 4. **Warning** pop up box: click **OK**
- C. **Restart**: use to Restart the Simulator. Use to restart after configuring or to reboot app.

4.2.3 Help menu



Help menu includes two submenus:

- **Simulator User Manual**: open the Simulator User Manual to use.
- **About BroadPOS Terminal Simulator**



4.3 Screen Area



App Screen Functionality:

Simulator zooms out to adjust the view, but will not zoom in.

4.4 Keyboard Area

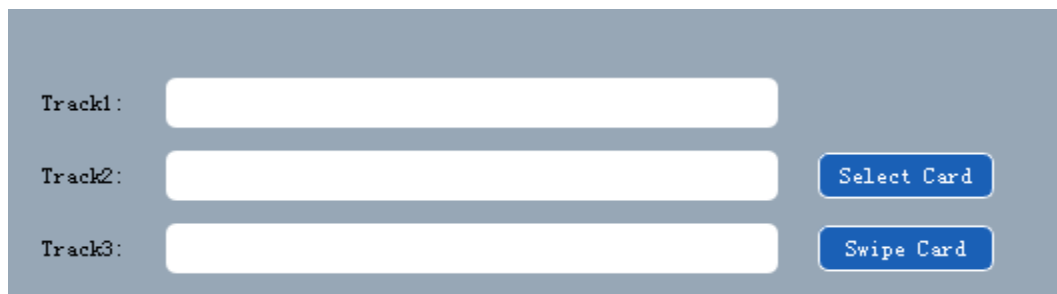


Keyboard Area Functionality:

- A. Simulator differs from POS terminal: Simulator users cannot click two buttons at the same time, so Simulator provides F1 – F4 keys.
 - F1 = **FUNC + 1** button
 - F2 = **FUNC + 2** button
 - F3 = **FUNC + 3** button
 - F4 = **FUNC + 4** button
- B. To switch letters, press the key continuously as well as pressing the alpha key.

4.5 Swipe Card Area

Figure: Input MagCard Info



Input MagCard Functionality:

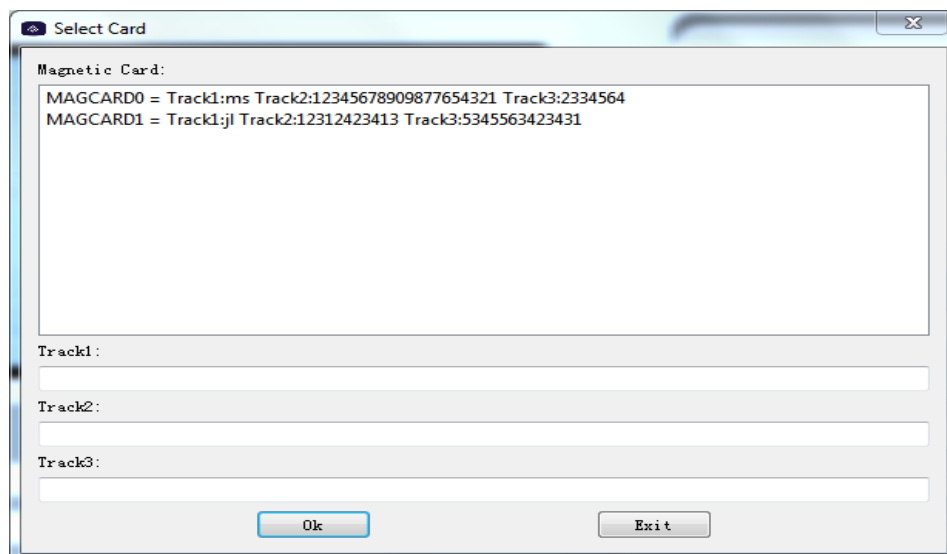
1. **Track1**, **Track2**, and **Track3** fields: enter card information directly into these fields.
2. Click **SwipeCard**

-or-

1. If user previously saved the card information in advance > click **SelectCard** > select

applicable card in the popup box.

Figure: Select MagCard Info



Select Msr Track Dialog Box Functionality:

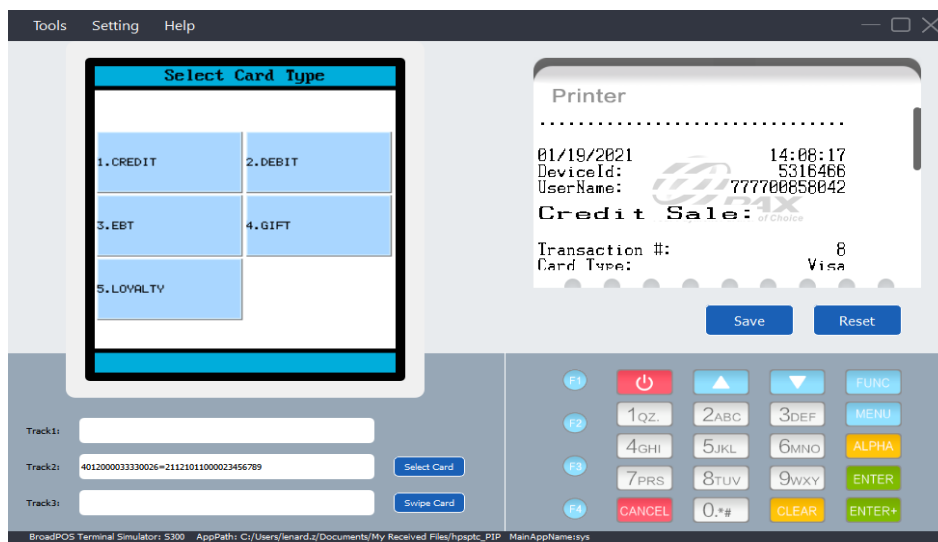
1. Click a card line item > click **Ok** to select card.

-or-

2. Double click a card line item to select card.

4.6 Printer area

Figure: Printer area



Printer Field Functionality:

- A. **Printer** field displays a virtual receipt to simulate the contents of an actual printed receipt.
- B. Click **Reset** button to clear **Printer** field.

5 FAQ

5.1 App does not work properly after installation

1. Delete the stored files generated by the Simulator during transaction simulations.
 - File path: C:\Users\UserName\AppData\Roaming\PAX\BroadPOS Terminal Simulator.
 - **Note:** UserName is PC user's name.
2. Reinstall Simulator after file deletion.

5.2 App does not display after load

Smaller Screen Size:

The configured screen size is **smaller** than the actual size. Screen displays a white area in the middle of screen.

Larger Screen Size:

The configured screen size is **larger** than the actual size. Screen displays with some white area.

6 Appendix

6.1 Terminal Simulator Screen Size

Model	Screen Size
Px7	800 x 440
Px5	800 x 440
MT30s	480 x 272
S80	128 x 64
S90	128 x 64

SP30s	320 x 200
S300	240 x 280
S500	320 x 200
S920	240 x 280
Q20	480 x 272
Q30	800 x 440
Q80	480 x 760
D210s	320 x 200
D200	320 x 200
D220	480 x 760
IM300	320 x 240
D190	320 x 216
Q25	320 x 216
Q25L	320 x 216
D190L	320 x 216
IM300L	320 x 216
IM310L	480 x 272
IM10	320 x 240
IM20	480 x 272

6.2 Default Password

The default password for all models is 1.